Fiscal Year:	FY 2016	**Task Last Updated:**	FY 09/21/2016
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	Comparative Analysis of Charged Particle-Induced Autosomal Mutagenesis in Murine Tissue and Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	03/14/2012	**End Date:**	03/31/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	4
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 3/31/2016 per S. Monk/LaRC (Ed., 9/14/15)		
Key Personnel Changes/Previous PI:			
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Task Description:	Astronauts in space are exposed to a variety of charged particle radiations, some of which travel at very high velocities that can penetrate spacecraft, damage tissues, and increase the risk of cancer. This proposal addresses two cancer development processes, genomic instability and aneuploidy, links them to each other and to specific mechanisms of mutagenesis at an endogenous autosomal locus in epithelial cells. The aims proposed use a comparative biology approach and a transgenic mouse model that involves both in vitro and in situ approaches. This strategy enables quantitative, coupled analyses of early changes in cells that survive the exposure and a direct test of the relevance of these changes to the development of persistent changes in situ. The studies address the radiation quality dependence of genomic instability and aneuploidy focusing on charged particle beams spanning a critical LET (linear energy transfer) range from 25 keV/µm to 151 keV/µm, and will define the shape of the dose-response for these endpoints, particularly at low fluence. The rationale for working in this range of LET is the observation of increased biological effectiveness for cancer relevant endpoints in our previous studies, including mutagenesis in human and rodent cells, and work by others concerning Harderian gland tumorigenesis in mice. The investigations test the hypothesis that the induction of genomic instability and aneuploidy will increase as a function of ionization density for low fluence exposures, leading to genome-wide alterations in viable, mutant cells found in normal tissue. This project will define links between the processes that lead to early establishment of genomic instability, aneuploidy, and mechanisms of autosomal mutagenesis found in many kinds of human epithelial cancers. These quantitative, mechanism-based studies will aid in the extrapolation of results across species to help reduce the uncertainties in cancer risk projections for astronauts exposed to spaceflight radiations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research examines the effects of charged particle radiations with different ionization densities using normal mammalian epithelium. The research is designed to help determine links between cancer development processes at relatively low doses. This work is of relevance to naturally occurring Earth-based exposures to densely ionizing radiations such as radon. It will also inform our understanding of the mechanisms underlying potential risks to normal tissues associated with the implementation of charged particle radiation therapy.
Task Progress:	During the final year of funding we focused on analysis of Oxygen ion induced mutants (250 MeV/amu, LET = 25 keV/µm. Results for 49 Aprt mutants arising following in vitro exposure to 1.6 Gy of O ions show that many exhibit radiation-signature mutations, including apparent mitotic recombination, multilocus deletion, and discontinuous loss of heterozygosity (LOH). Small intragenic events were also found in this cohort, and at a higher proportion than in cohorts isolated after exposure to Si ions or Fe ions. Such small events were more common following exposure to sparsely ionizing protons (1 GeV). Taken together, there tends to be a trend toward larger, multilocus events as LET increases. Intragenic events were also prevalent in primary kidney clones isolated after in situ exposure to a low dose of O ions (0.4 Gy), while deletions or discontinuous LOH events were rare. However, there is evidence from analyses involving other chromosomes that O ions are capable of producing genome-wide rearrangements, similar to what we have observed for heavier ions. Additional molecular analysis is ongoing. In this final year, we also addressed three additional goals. The first was to investigate the effect of dose-fractionation on genotoxic endpoints in cultured mouse kidney epithelial cells. We examined cytotoxicity and Aprt mutation induction following an acute dose of Si ions (240 MeV/amu, LET = 78 keV/µm) or the same total dose administered in 4 daily fractions. The study covered a dose range from 0-88 cGy. For the fractionated study, the lower doses/fraction were 2.75 cGy and 5.5 cGy/day. At these lower doses, the dose per fraction corresponded to either 0 or 1 particle traversal of a given cell nucleus. The probabilities of >1 traversal by the central region of a Si ion track were extremely low. Dose fractionation was not sparing with regard to cytotoxicity at any of the doses examined. Furthermore, there was no sparing from the induction of Aprt mutants. Taken together, the results indicate that the effects of heavy ion exposure are additive for the two endpoints examined in this model epithelium. The second additional goal was to develop new technological approaches to the characterization of genome-wide events following exposure to charged particle radiation, e.g., a low to moderate fluence of Fe ions. Earlier, we used PCR-based assays to follow the loss of heterozygosity (LOH) along mouse chromosome 8 using 13 markers arrayed along the chromosome. We also used a similar approach to examine genome-wide changes in Aprt mutant colonies and in non-mutant clones. Here we used single-nucleotide polymorphism (SNP) analyses of the whole genome to examine a series of Aprt mutant and non-mutant samples that had been exposed to 0.25, 0.5, or 1 Gy of 1 GeV/amu Fe ions (lowest dose corresponds to an average of approximately 1 particle track/cell nucleus). DNAs were subjected to comprehensive SNP analysis by a commercial vendor using an updated set of probes surveying 100,000 murine SNPs. Roughly 23,000 of these SNPs were expected to be heterozygous on the B6D2F1 strain background. Charts of individual chromosomes were devised to examine the extent of LOH in each of the DNAs submitted for analysis. The results were generally in good agreement with the previous PCR-based analysis, but the level of coverage of the genome is much greater using the SNP approach. The third additional goal was to implement a detailed breeding approach to develop a mouse strain to enable a linkage between autosomal mutagenesis and the induction of kidney tumors. The plan was to devise a strain based on the B6D2F1 mouse model, where we have accumulated extensive knowledge regarding the quantitative and molecular aspects of charged particle-induced mutations in kidney epithelium. The strategy included the use of a conditional allele for one of the genes that is known to predispose people to kidney cancer. The breeding strategy was successful.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
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